

Fady Abousifein

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EDUCATION

McMaster University

Hamilton, ON

Mechatronics Engineering CO-OP

Skills: C, C++, Python, Linux OS, Matlab, Ladder Logic | Mitsubishi GX Works2, GT Designer3, Simulink, LTSpice, MultiSim | CATIA V6, SolidWorks, AutoCAD Electrical, DraftSight | Arduino, STM32, Raspberry Pi, PLCs, FPGAs | Oscilloscope, Soldering

EXPERIENCE

Electrical Engineering Intern

May 2026 – Present

HONDA of Canada Manufacturing (HCM)

Alliston, ON

- Developed Mitsubishi PLC ladder logic and HMI applications for HCM's first new model engine launch in 10+ years, enabling multi-model engine head production using **Mitsubishi GX Works2** and **GT Designer3**.
- Developed an **industrial sensor-based model detection system** integrating PLC ladder logic, HMI software, and electrical hardware to automate engine model identification and enable multi-model manufacturing.
- Designed and validated a custom mechanical fixture in **CATIA** for multi-model engine head hardness testing, prototyping and optimizing the design for manufacturability and production deployment.
- Reverse-engineered legacy systems by analyzing PLC logic, HMI interfaces, electrical schematics, and robot programs, implementing controls improvements on live **Yaskawa Motoman** and **FANUC robotic systems**.

Hardware Design Engineering Intern

May 2025 – Aug 2025

Skyjack

Guelph, ON

- Reverse-engineered a competitor's collision detection system for scissor lifts using **Arduino** and **ultrasonic sensing**. Implemented live distance measurements and automated lift control, preventing over \$100K in losses.
- Designed a custom **Arduino shield PCB** in **EasyEDA**. Implemented transistor based level shifting, power regulation, and transistor driver circuits to interface a 24V industrial scissor lift system with a 5V microcontroller.
- Developed a prototype combining **Arduino control, ultrasonic sensing, and CAN communication** with a 24V scissor lift. Developed electrical schematics using **DraftSight** and presented prototype to a major customer.
- Debugged and validated collision detection system logic using **Falstad Simulation**, multimeter measurements, and **PCAN-View** CAN analyzer to verify sensor I/O, controller logic, lift communication, verifying functionality

Combinatorial Algorithms Research Assistant

Jun 2023 – Sept 2023

McMaster University

Hamilton, ON

- Translated pseudocode from research papers into efficient **C++** and **C** algorithms enabling testing and validation of combinatorial optimization techniques on real-world datasets.
- Used **Maple** for mathematical analysis and validation of algorithms, supporting research experiments and performance comparisons.
- Contributed to the optimization of algorithms, reducing runtime in targeted cases and accelerating computational experiments.

Productive Software Development Research Assistant

Jun 2022 – Sept 2022

The University of Waterloo

Waterloo, ON

- Developed features for The Checker Framework VSCode extension, enhancing Java's type-checking workflows and code safety.
- Migrated project dependencies from **yarn** to **npm** to maintain compatibility with evolving **VSCode extension APIs**, improving dependency management and project stability.
- Updated and maintained technical documentation and project configuration to support future development and version updates.

PROJECTS

Embedded Stepper Motor Control System

Apr 2024

Embedded Systems Design Project

McMaster University, ON

- Engineered a real-time stepper motor control system on the **STM32F429ZI** using **mbed OS**, implementing non-blocking motion control with interrupt-based timing for precise angular positioning.
- Developed **interrupt-driven controls** for stepper motor direction, speed, and stepping mode, ensuring immediate responsiveness to user inputs through debounced hardware inputs.
- Designed and implemented a dynamic **LCD interface** to display user-specific information, motor speed, stepping mode, and direction, improving system usability and real-time feedback.